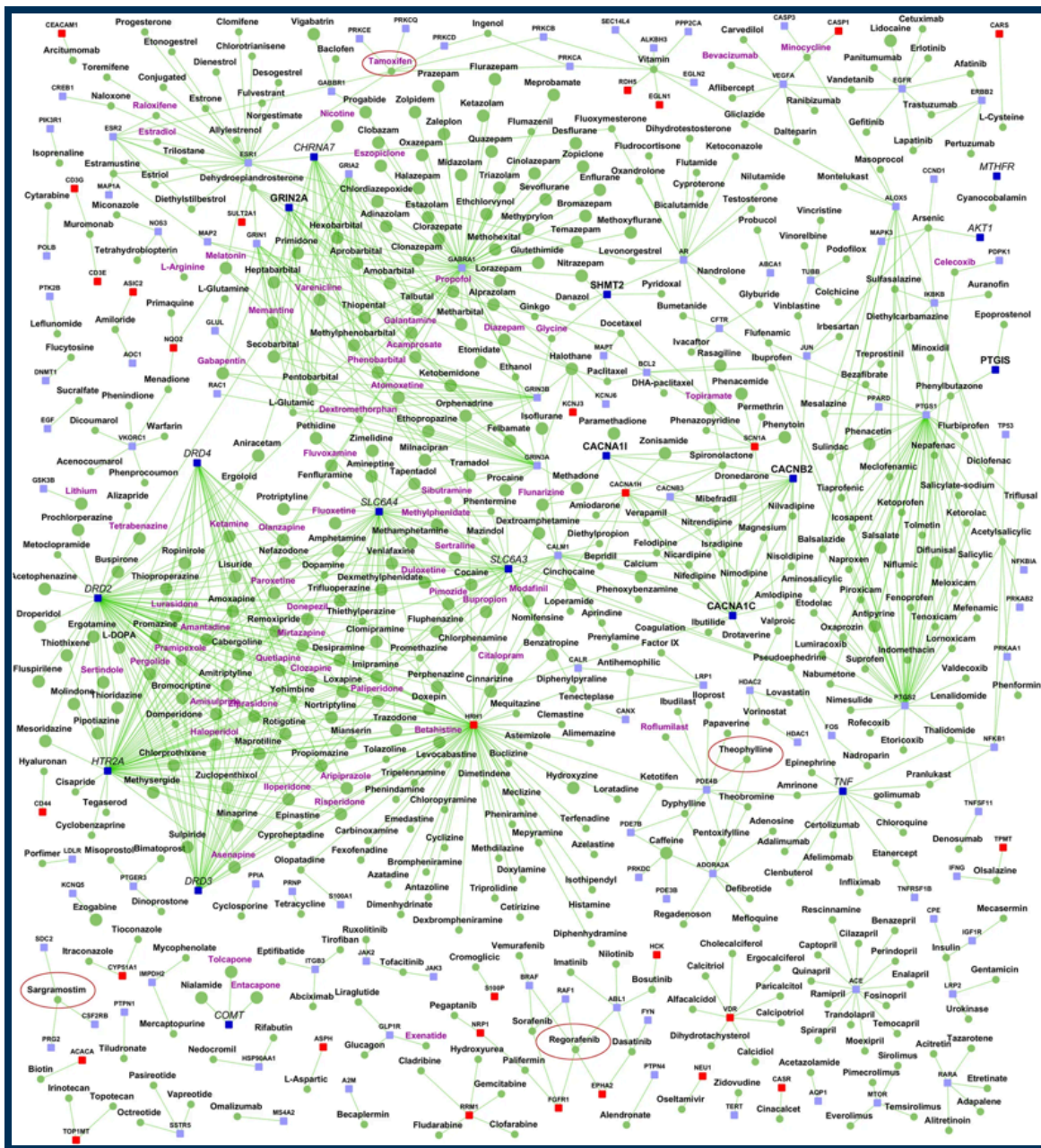


L1: Machine Learning and Network Science

We will also study problems at the intersection of Network Science and Machine Learning.

As you probably know, Machine Learning generates statistical models from data and then uses these models in classification, regression, clustering, and other similar tasks.

Network Science has contributed to this field by focusing on graph models – statistical models of static or dynamic networks that can capture the important properties of real-world networks in a parsimonious manner.



The image above comes from a recent research paper about schizophrenia. (Image Source: [Ganapathiraju, M., Thahir, M., Handen, A. et al. Schizophrenia interactome with 504 novel protein-protein interactions. npj Schizophr 2, 16012 \(2016\).](https://www.nature.com/articles/npschz201612) <https://www.nature.com/articles/npschz201612>.)

It shows the interactions between genes associated with schizophrenia, and drugs that target either specific genes/proteins or protein-protein interactions. Machine Learning models have been used to predict previously unknown interactions between drugs and genes.

The drugs are shown as round nodes in green, and genes as square nodes in dark blue, light blue or red. Nervous system drugs are shown as larger size green colored

nodes compared with other drugs. Drugs that are in clinical trials for schizophrenia are labeled purple. You can explore the visualization interactively with the following link:

[Schizophrenia interactome with 504 novel protein–protein interactions](https://www.nature.com/articles/npjschz201612/figures/6) 

[\(https://www.nature.com/articles/npjschz201612/figures/6\)](https://www.nature.com/articles/npjschz201612/figures/6)